

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

7th Semester of B. Pharm. Examination

University Theory Examination November 2016

PH403 MEDICINAL CHEMISTRY- III

Date: 18.11.2016 , Friday

Time: 10.00 a.m to 1.00 p.m

Maximum Marks: 80

Instructions:

1. Figures to right indicate marks.
2. Section wise separate answer book to be used.
3. Draw neat sketches wherever necessary.

SECTION- I

Q 1 Attempt any TWO of the following

- A Define QSAR. Classify various physicochemical properties studied by QSAR and explain any one in detail. **10**
- B Add a short note on following: **10**
1. Hansch analysis
 2. Pros and Cons of 3D QSAR
- C Calculate log P values for following compounds (Substituent Hydrophobicity Constant values for OH= -0.67, CH₃= 0.53, Cl= 0.71, CONH₂= -1.49; log P for benzene is 2.13): **10**
1. p-hydroxy toluene
 2. m-chlorobenzamide
 3. p-chloro toluene
 4. 3,4,5-trichloro phenol

Q 2 Attempt any TWO of the following

- A Draw and explain topliss scheme for aromatic substituents. **10**
- B Add a short note on following: **10**
1. HTS
 2. SOSA approach for drug discovery
- C Enumerate various types of biological screening methods useful for bioassay of drug like molecules and explain any three in brief. **10**

SECTION- II

Q 3 Attempt any TWO of the following

- A **05**
1. Write down SAR of thiazide diuretics.
 2. Explain chemistry, mechanism of action and uses of cardiac glycosides. **05**
- B Write a note on Structure Based Drug Design. **10**

- C 1. Outline synthesis of acetazolamide and acetyl salicylic acid. 05
 2. Write a note on antianginal agents. 05
- Q 4 Attempt any TWO of the following**
- A Define and classify antihypertensive drugs giving one structure in each category. Explain SAR of 1,4-DHP derivatives in detail. 10
- B 1. Write SAR and mechanism of action of HMG-CoA reductase inhibitors. 05
 2. Write a brief note on drugs used to treat arrhythmias. 05
- C 1. Explain various types of docking methods. 05
 2. Write a note on Denovo drug design. 05
-

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
Seventh Semester of B. Pharm. Examination
University Theory Examination December 2016
PH404 Pharmaceutical Analysis-III

Date: 05.12.2016, Monday

Time: 10.00 a.m to 01.00 p.m

Maximum Marks: 80

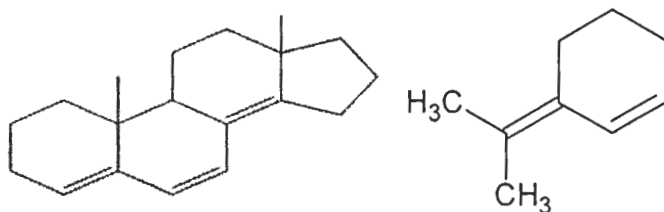
Instructions:

1. Figures to right indicate marks.
2. Section wise separate answer book to be used.
3. Draw neat sketches wherever necessary.

SECTION- I

Q 1 Attempt any **TWO** of the following.

- A** (1) Write a note on factors affecting the $\lambda_{\max}$ of a compound in a UV region. 05
(2) Calculate $\lambda_{\max}$ of following structures. 05



- B** (1) Explain the principle of FT-IR spectroscopy with suitable diagram of the instrumentation and its advantage over dispersive IR spectroscopic technique. 05
(2) Enlist the types of UV spectrophotometers and explain double beam in Space with a suitable diagram. 05

- C** (1) Write a note on factors affecting fluorescence intensity. 05
(2) Write a note on ATR. 05

Q 2 Attempt any **TWO** of the following.

- A** Write a short note on UV detectors. 10
- B** (1) Explain the sample handling techniques of IR Spectroscopy. 05
(2) Define specific absorbance, chromophore, auxochrome, phosphorescence and wavenumber. 05
- C** (1) Explain principle and instrumentation of Raman spectroscopy. 05
(2) Comment on these statements: 05
a) Doublet is seen in IR spectra of aldehyde containing compound
b) Different types of amines show different pattern in the functional group region in the IR Spectra.

SECTION- II

- Q 3** Attempt any **TWO** of the following.
- A** (1) Define column chromatography. Give detailed classification of column chromatography. Explain retention time and resolution. **05**
(2) Write a note on mobile phases and stationary phases used in HPLC. **05**
- B** (1) Draw Schematic diagram of HPLC. Write short note on pumps used in HPLC. **05**
(2) Discuss van Deemter equation. **05**
- C** (1) Differentiate HPTLC from TLC. **02**
(2) Outline basic components and steps for HPTLC technique. **02**
(3) Describe Silica gel GF₂₅₄. **02**
(4) How sample is applied in HPTLC technique? Explain. **02**
(5) Enlist applications of HPTLC. **02**
- Q 4** Attempt any **TWO** of the following.
- A** Write ideal requirements for HPLC detectors. Describe detectors commonly used in HPLC. **10**
- B** (1) What are the limitations of GC? Write about its principle and applications. **05**
(2) Classify GC detectors. Write about design and features of TCD. **05**
- C** (1) Give a brief note on carrier gases and columns used in GC. **05**
(2) Write about separation principle and applications of Ion Exchange Chromatography. **05**
-